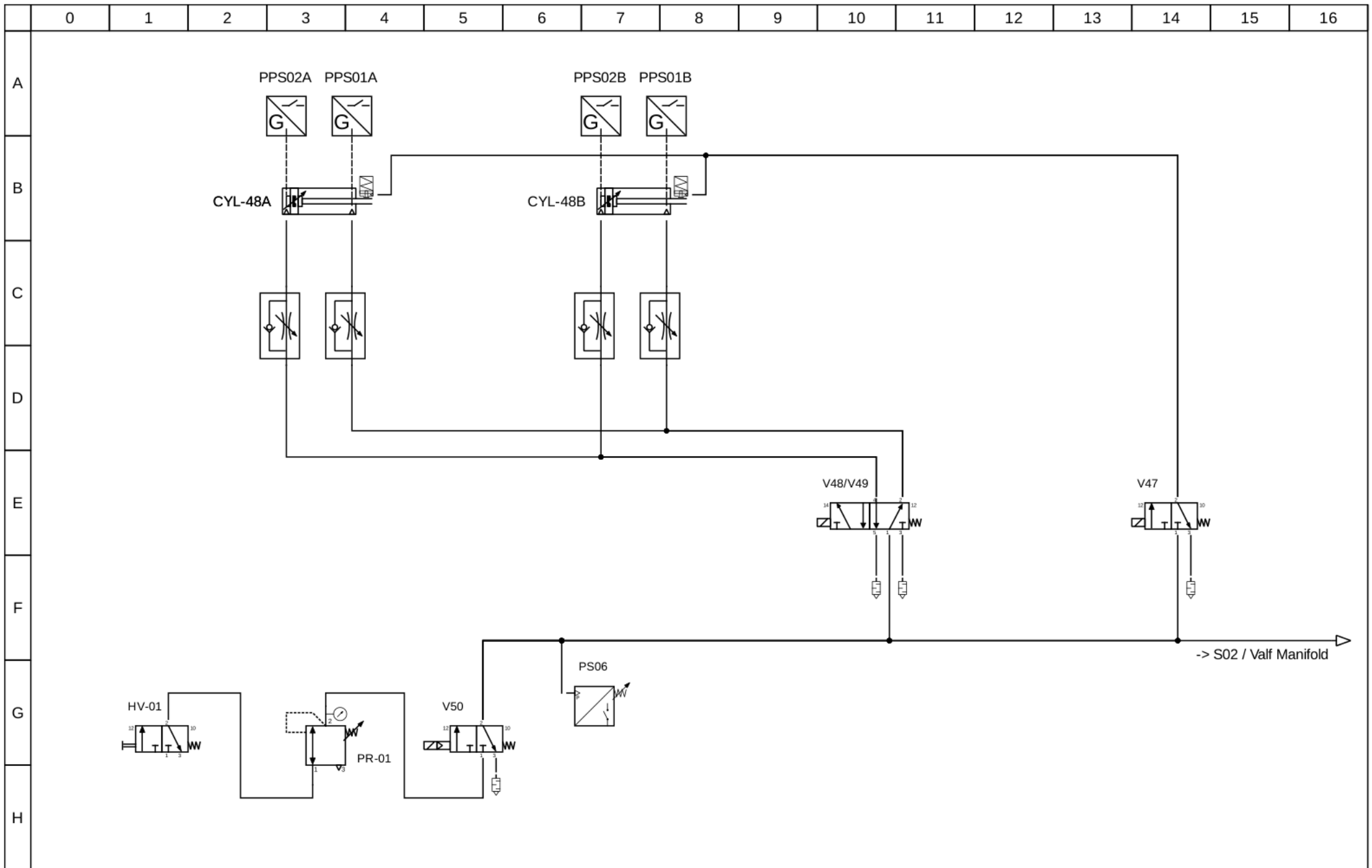


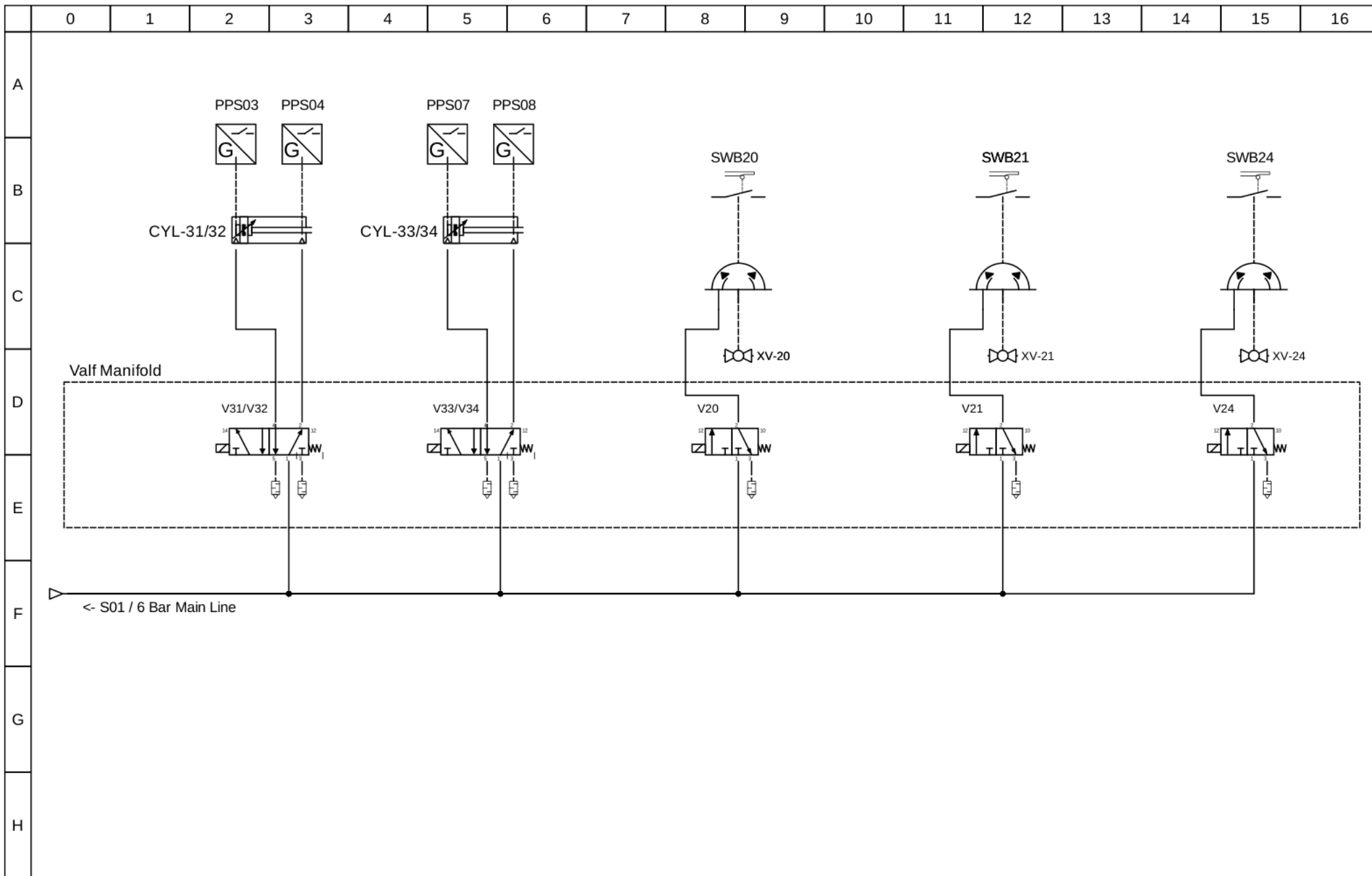
1326050-KSYSTEM-KBN 2B 1650 PNEUMATIC PLAN

Issue Date : 01.10.2026 / Prepared by : Fatih GÜRAL

DolfinTM
INDUSTRIAL PARTS WASHING MACHINES

Washing Systems...





1 — Function overview

Industrial parts washer KBN 2B 1650. Air is prepared on the main line; directional valves drive cylinders and quarter-turn actuators for drain / fill / cover motions.

AIR PREP

HV-01 isolation → PR-01 filter-regulator (~6 bar) → V50 soft-start → PS06 pressure switch.

5/2 VALVES

One coil = one direction, other coil = reverse. V31/V32 and V33/V34 each drive one cylinder.

3/2 + XV

V20 / V21 / V24 energize spring-return CONVALVE actuators (XV). SWB confirms open/closed.

FEEDBACK

PPS (CMSE-07FY) = cylinder ends. SWB/API200 = actuator box switches.

2 — Tag relationship map

PNEUMATIC TAG	DRIVES / SENSES	ROLE
V31/V32	CYL-31/32 + PPS	Wash drain piston OPEN/CLOSE (5/2)
V33/V34	CYL-33/34 + PPS	Rinse drain piston OPEN/CLOSE (5/2)
V20 / V21 / V24	XV-20/21/24 + SWB	Fill / cover quarter-turn actuators (3/2)
V50	Main air soft-start	Soft start upstream of consumers
V47	3/2 auxiliary	Single-solenoid consumer
V48/V49	5/3 single body	Dual coil exhaust-center
PPS...	CYL magnet grooves	Piston reed sensors
SWB20/21/24	On XV actuators	OPEN/CLOSED mechanical switches

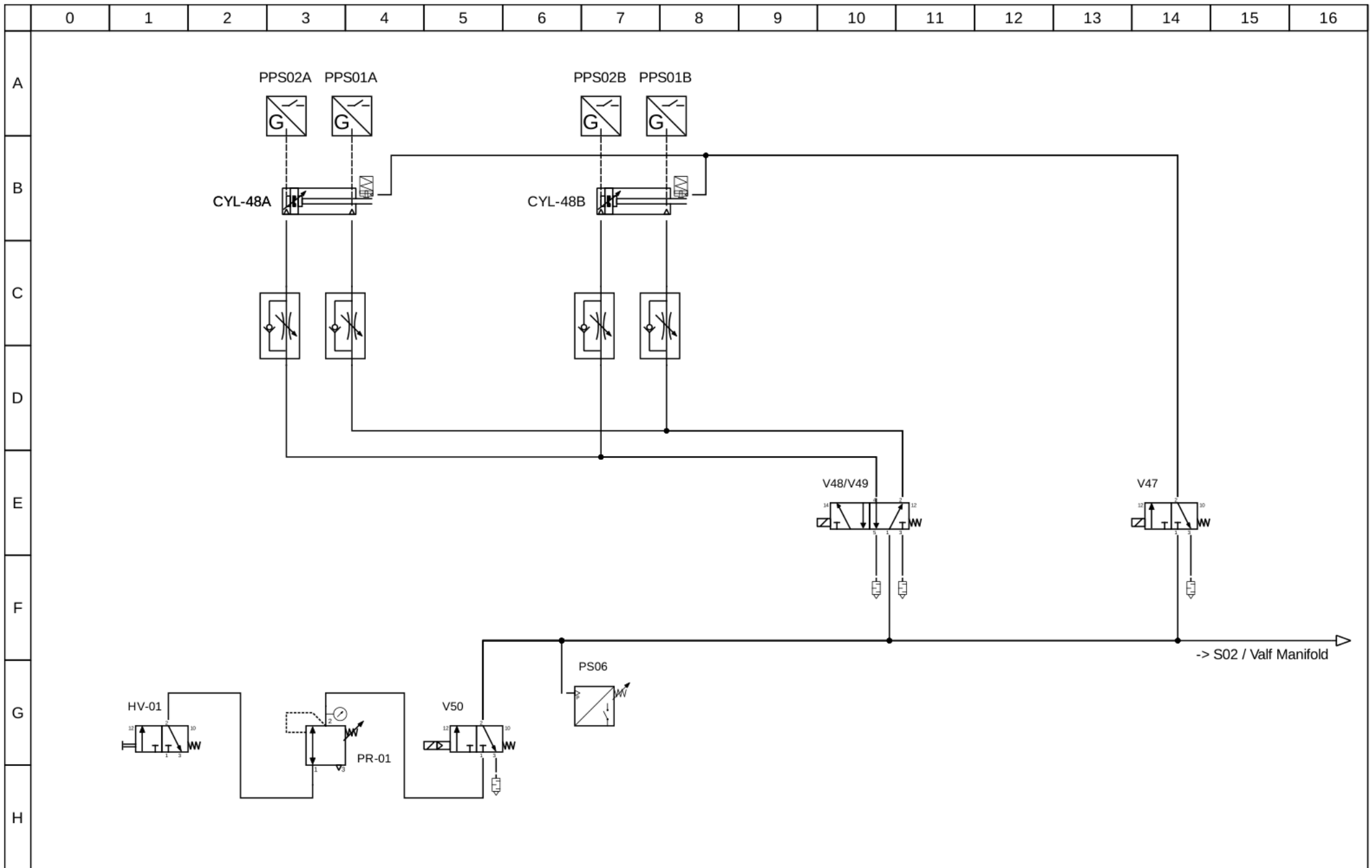
3 — Electrical I/O (valves & actuators)

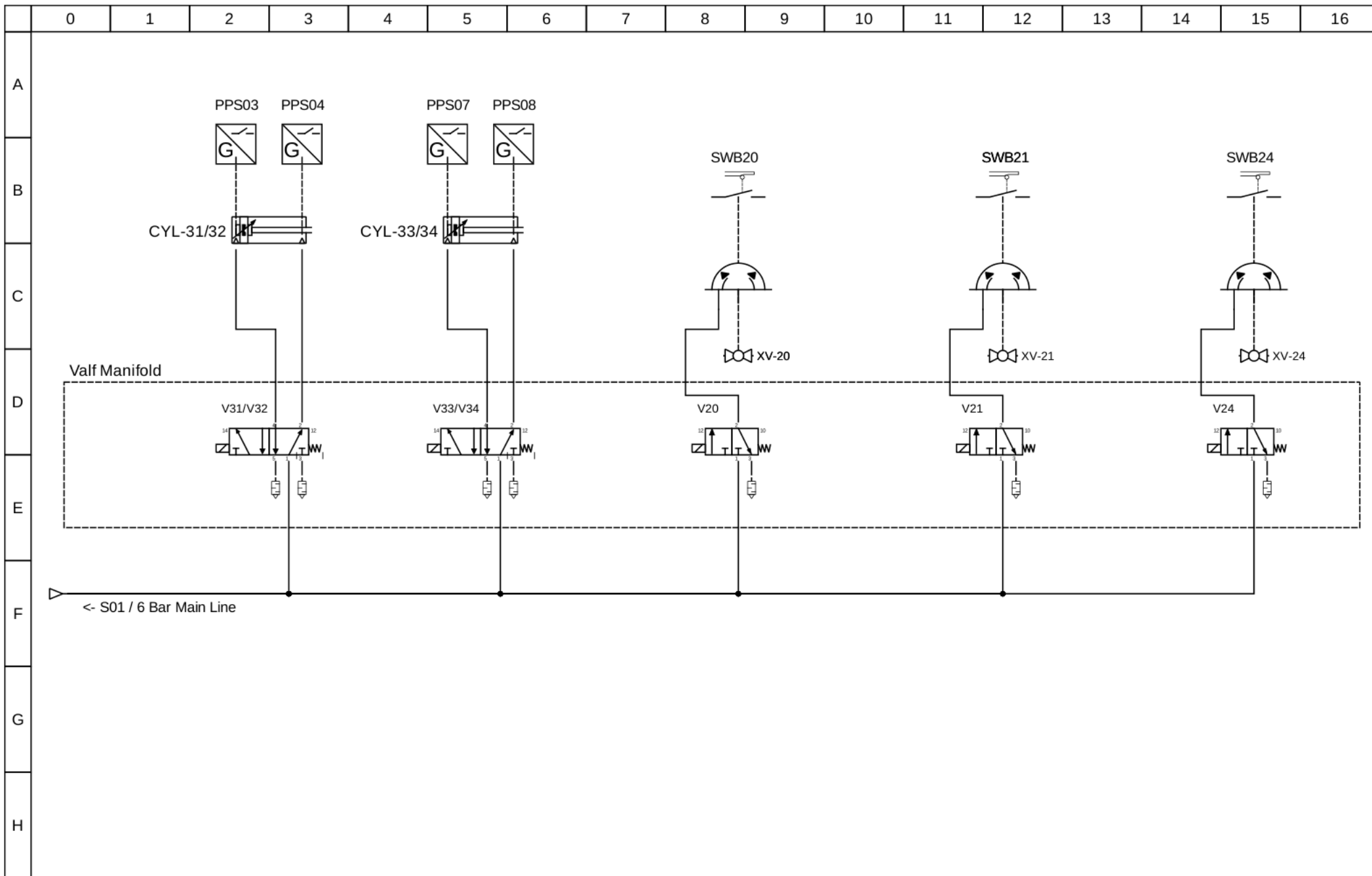
Omron CP2E · G9PS safety I/O · valve & actuator mapping by process function

PLC / SAFETY I/O	ELECTRICAL FUNCTION	PNEUMATIC LINK (BY FUNCTION)
Q1.1	WASH DRAIN VALVE OPEN	V31/V32 coil A → CYL-31/32
Q1.2	WASH DRAIN VALVE CLOSE	V31/V32 coil B → CYL-31/32
Q1.3	RINSE DRAIN VALVE OPEN	V33/V34 coil A → CYL-33/34
Q1.4	RINSE DRAIN VALVE CLOSE	V33/V34 coil B → CYL-33/34
Q2.0	WASH AUTO FILL VALVE	3/2 fill valve (V20/V21/V24 group)
Q2.1	RINSE AUTO FILL VALVE	3/2 fill valve (V20/V21/V24 group)
G9PS So0	COVER OPEN VALVE	Cover XV OPEN
G9PS So1	COVER CLOSE VALVE	Cover XV CLOSE
G9PS So*	MAIN VALVE	Main process / air valve
I1.3 / I1.4	WASH DRAIN PISTON OPEN / CLOSED	PPS on CYL-31/32
I1.5 / I1.6	RINSE DRAIN PISTON OPEN / CLOSED	PPS on CYL-33/34
I2.3 / I2.4	WASH AUTO FILL SW BOX OPEN / CLOSED	SWB on fill actuator
I2.5 / ...	RINSE AUTO FILL SW BOX OPEN / CLOSED	SWB on fill actuator
I2.6 / I2.7	COVER PISTON OPEN / CLOSED SENSOR	Cover position feedback
I1.9 (via G9PS)	COVER CLOSED SIGNAL	Cover closed interlock

#	TAG	COMPONENT TYPE	FOLIO	QTY	PRODUCT / PHOTO
S01 — AIR PREPARATION / MAIN LINE					
1	PR-01	Pressure regulator / Filter-regulator	1/2	1	 AW40-F04CE-D FILTER REGULATOR SMC
2	HV-01	3/2 manual valve (pressure relief)	1/2	1	 VHS40-F04-D PRESSURE RELIEF 3 PORT VALVE SMC
3	PS06	Modular pressure switch	1/2	1	 IS10E-40-F04-6L-D PRESSURE SWITCH MODULAR SMC
S01 — VALVES					
4	V50	Soft / slow start valve	1/2	1	 AV4000-F04-5DZB-A SOFT START VALVE 1/2-24VDC SMC
5	V47	3/2 single-solenoid valve	1/2	1	 AIRTAC 3V210-08 3/2 Single Solenoid · 1/4" AIRTAC
6	V48/V49	5/3 double-solenoid valve (single body)	1/2	1	 AIRTAC 4V230E-08 5/3 Exhaust Center · 1/4" AIRTAC
S01 — ACTUATORS & SENSORS					
7	CYL-48A / CYL-48B	Locking double-acting cylinder	1/2	2	 CP96NDB50-1200C CYLINDER WITH LOCK Ø50 STR=1200 SMC
8	PPS01A / PPS01B / PPS02A / PPS02B	Piston magnetic sensor (reed)	1/2	4	 AIRTAC CMSE-07FY CMS Series · E groove · flying lead AIRTAC
S02 — VALVE MANIFOLD					
9	V20 / V21 / V24	3/2 single-solenoid valve	2/2	3	 AIRTAC 3V210-08 3/2 Single Solenoid · 1/4" AIRTAC
10	V31/V32 + V33/V34	5/2 double-solenoid valve	2/2	2	 AIRTAC 4V220-08 5/2 Double Solenoid · 1/4" AIRTAC

#	TAG	COMPONENT TYPE	FOLIO	QTY	PRODUCT / PHOTO
S02 — ACTUATORS					
11	CYL-31/32 + CYL-33/34	Double-acting magnetic profile cylinder Ø80×100	2/2	2	 Luftec LDNT80X100-S Magnetic cushioned pneumatic cylinder Luftec
12	XV-20 / XV-21 / XV-24	Single-acting rotary actuator (spring return)	2/2	3	 CONVALVE PAC Quarter-turn pneumatic actuator · spring return CONVALVE
S02 — SENSORS / SWITCHES					
13	SWB20 / SWB21 / SWB24	Actuator position indicator / limit switch box	2/2	3	 CONVALVE API200 ACTUATOR POSITION INDICATOR · 2×SPDT · IP67 CONVALVE
14	PPS03 / PPS04 / PPS07 / PPS08	Piston magnetic sensor (reed)	2/2	4	 AIRTAC CMSE-07FY CMS Series · E groove · flying lead AIRTAC
UNLABELED / GENERAL + QUOTATION EXTRAS					
15	—	Silencer 3/8	1-2	1	 AN30-03 SILENCER 3/8 SMC
16	—	Silencer 1/2	1-2	1	 AN40-04 SILENCER 1/2 SMC
17	—	Modular spacer / spare part	—	2	 Y400T-D SPARE PART SMC
18	—	Modular spacer / spare part	—	1	 Y400-D SPARE PART SMC
SUMMARY — PISTON SENSORS					
19	PPS (all)	Piston magnetic sensor — single type	1-2	8	 AIRTAC CMSE-07FY All PPS items same product AIRTAC





1 — Funktionsübersicht

Industrielle Teilewaschanlage KBN 2B 1650. Druckluft wird aufbereitet; Wegeventile steuern Zylinder und Schwenkantriebe für Ablauf / Füllung / Deckel.

LUFTAUFBEREITUNG

HV-01 Absperrung → PR-01 Filterregler (~6 bar) → V50 Soft-Start → PS06 Druckschalter.

5/2-VENTILE

Eine Spule = eine Richtung, andere Spule = Gegenrichtung. V31/V32 und V33/V34 je ein Zylinder.

3/2 + XV

V20 / V21 / V24 speisen CONVALVE-Schwenkantriebe (XV). SWB meldet Offen/Geschlossen.

RÜCKMELDUNG

PPS (CMSE-07FY) = Zylinderendlagen. SWB/API200 = Schalterbox.

2 — Kennzeichen-Zuordnung

PNEUMATIK-KENNUNG	TREIBT / ERFASST	FUNKTION
V31/V32	CYL-31/32 + PPS	Waschablauf-Kolben ÖFFNEN/SCHLIESSEN (5/2)
V33/V34	CYL-33/34 + PPS	Spülablauf-Kolben ÖFFNEN/SCHLIESSEN (5/2)
V20 / V21 / V24	XV-20/21/24 + SWB	Füll- / Deckel-Schwenkantriebe (3/2)
V50	Hauptluft Soft-Start	Sanftanlauf vor Verbrauchern
V47	3/2 Hilfsventil	Einfachmagnet-Verbraucher
V48/V49	5/3 ein Gehäuse	Doppelspule Mittelabluft
PPS...	CYL-Magnetnuten	Kolben-Reed-Sensoren
SWB20/21/24	An XV-Antrieben	OPEN/CLOSED Endschalter

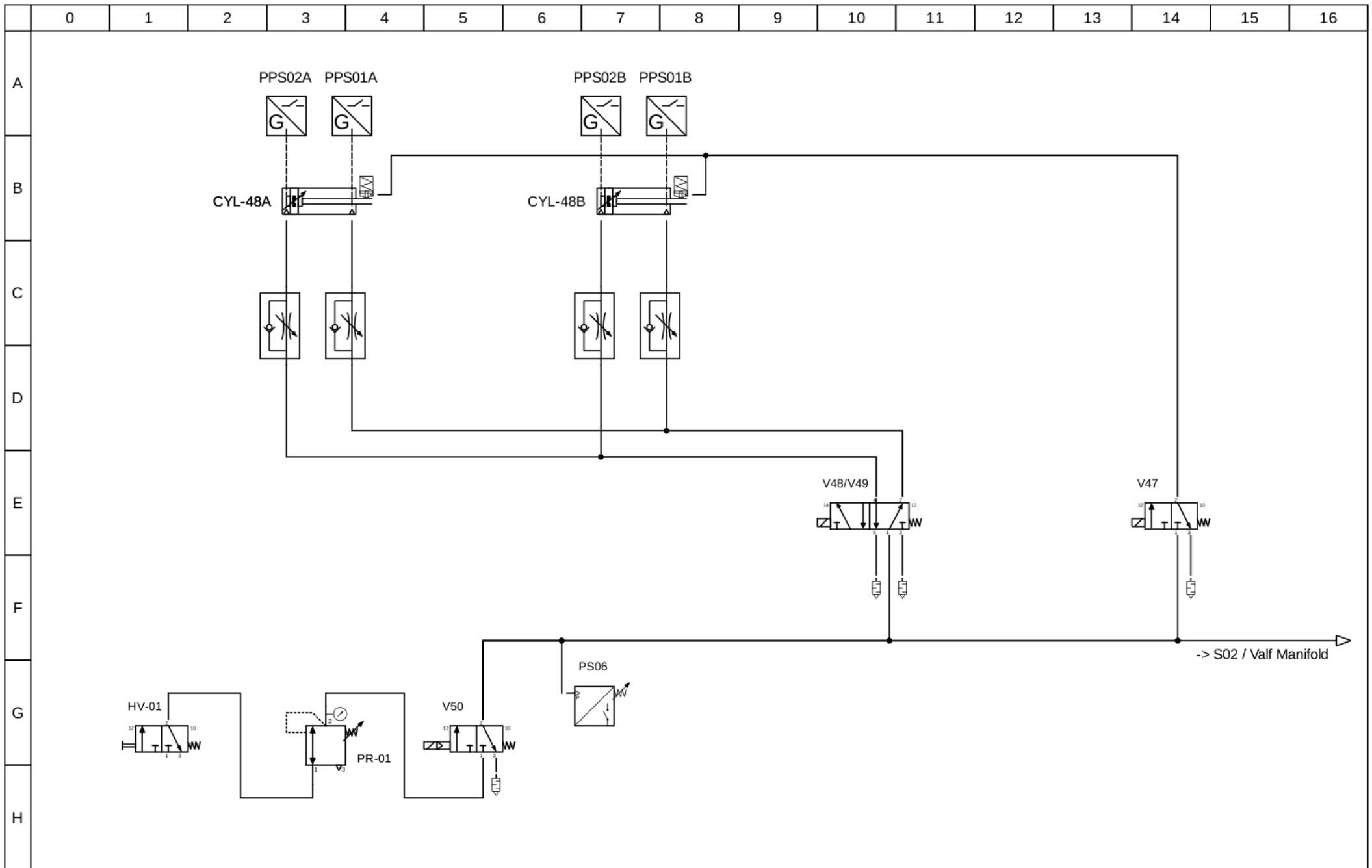
3 — Elektrische E/A (Ventile & Antriebe)

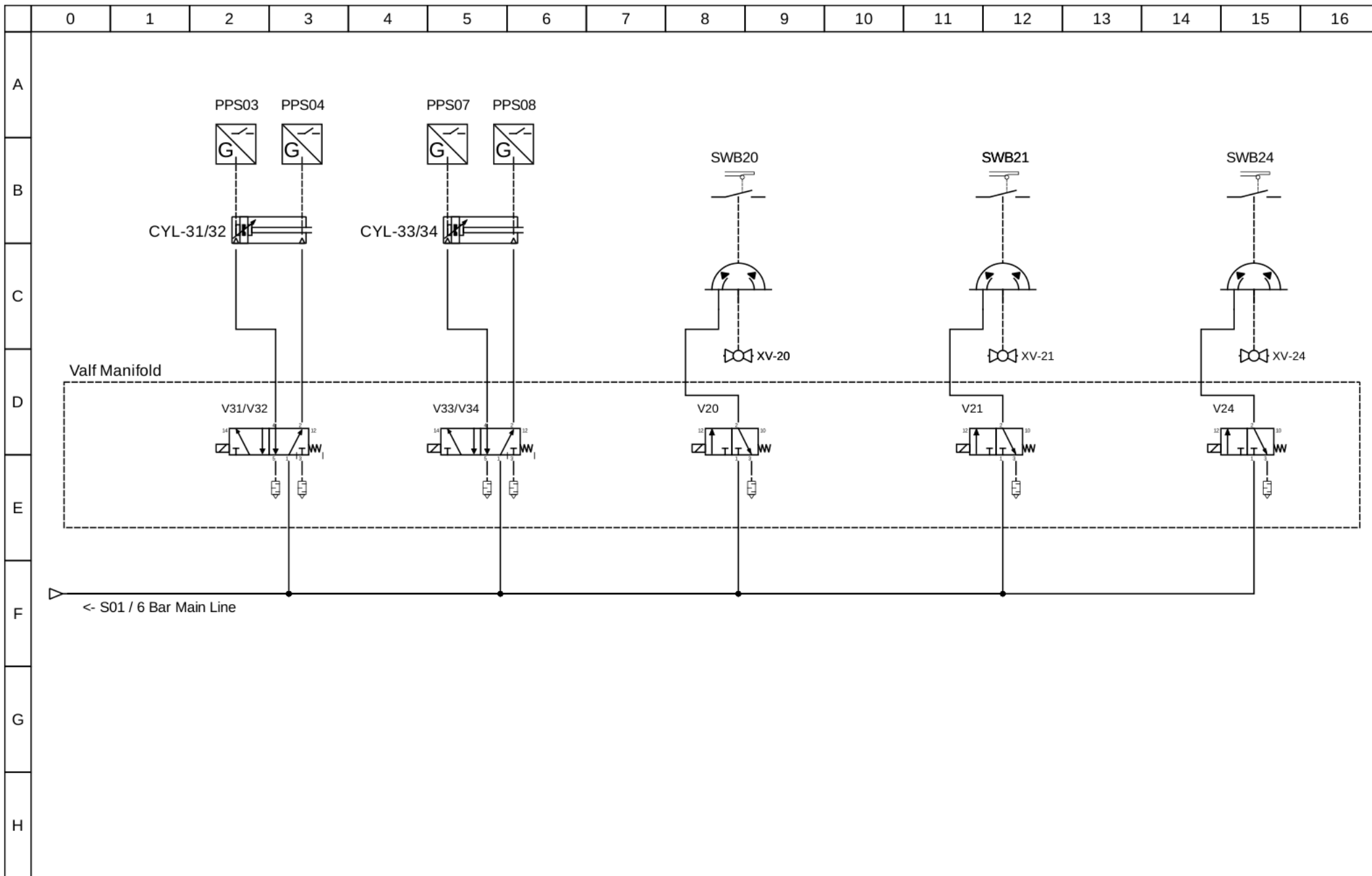
Omron CP2E · G9PS Safety-E/A · Ventil- & Antriebszuordnung nach Prozessfunktion

SPS / SAFETY E/A	ELEKTRISCHE FUNKTION	PNEUMATIK-BEZUG
Q1.1	WASCHABLAUFVENTIL ÖFFNEN	V31/V32 Spule A → CYL-31/32
Q1.2	WASCHABLAUFVENTIL SCHLIESSEN	V31/V32 Spule B → CYL-31/32
Q1.3	SPÜLABLAUFVENTIL ÖFFNEN	V33/V34 Spule A → CYL-33/34
Q1.4	SPÜLABLAUFVENTIL SCHLIESSEN	V33/V34 Spule B → CYL-33/34
Q2.0	WASCH AUTO. FÜLLVENTIL	3/2 Füllventil (V20/V21/V24)
Q2.1	SPÜL AUTO. FÜLLVENTIL	3/2 Füllventil (V20/V21/V24)
G9PS So0	COVER OPEN / Deckel öffnen	Deckel-XV ÖFFNEN
G9PS So1	COVER CLOSE / Deckel schließen	Deckel-XV SCHLIESSEN
G9PS So*	MAIN VALVE / Hauptventil	Hauptprozessventil
I1.3 / I1.4	WASCHABLAUF-KOLBEN OFFEN / ZU	PPS an CYL-31/32
I1.5 / I1.6	SPÜLABLAUF-KOLBEN OFFEN / ZU	PPS an CYL-33/34
I2.3 / I2.4	WASCH AUTO FILL SW BOX OFFEN / ZU	SWB am Füllantrieb
I2.5 / ...	SPÜL AUTO FILL SW BOX OFFEN / ZU	SWB am Füllantrieb
I2.6 / I2.7	COVER PISTON OPEN / CLOSED	Deckel-Rückmeldung
I1.9 (via G9PS)	COVER CLOSED SIGNAL	Deckel-Zu-Verriegelung

#	KENNUNG	KOMPONENTENTYP	FOLIO	STK.	PRODUKT / FOTO
S01 — LUFTAUFBEREITUNG / HAUPTLEITUNG					
1	PR-01	Druckregler / Filterregler	1/2	1	 AW40-F04CE-D FILTER REGULATOR SMC
2	HV-01	3/2 Handventil (Druckentlastung)	1/2	1	 VHS40-F04-D PRESSURE RELIEF 3 PORT VALVE SMC
3	PS06	Modularer Druckschalter	1/2	1	 IS10E-40-F04-6L-D PRESSURE SWITCH MODULAR SMC
S01 — VENTILE					
4	V50	Sanftanlaufventil (Soft-Start)	1/2	1	 AV4000-F04-5DZB-A SOFT START VALVE 1/2-24VDC SMC
5	V47	3/2 Magnetventil (einfach)	1/2	1	 AIRTAC 3V210-08 3/2 Single Solenoid · 1/4" AIRTAC
6	V48/V49	5/3 Doppelmagnetventil (ein Gehäuse)	1/2	1	 AIRTAC 4V230E-08 5/3 Exhaust Center · 1/4" AIRTAC
S01 — ANTRIEBE & SENSOREN					
7	CYL-48A / CYL-48B	Doppelwirkender Zylinder mit Verriegelung	1/2	2	 CP96NDB50-1200C CYLINDER WITH LOCK Ø50 STR=1200 SMC
8	PPS01A / PPS01B / PPS02A / PPS02B	Kolben-Magnetsensor (Reed)	1/2	4	 AIRTAC CMSE-07FY CMS Series · E groove · flying lead AIRTAC
S02 — VENTILINSEL					
9	V20 / V21 / V24	3/2 Magnetventil (einfach)	2/2	3	 AIRTAC 3V210-08 3/2 Single Solenoid · 1/4" AIRTAC
10	V31/V32 + V33/V34	5/2 Doppelmagnetventil	2/2	2	 AIRTAC 4V220-08 5/2 Double Solenoid · 1/4" AIRTAC

#	KENNUNG	KOMPONENTENTYP	FOLIO	STK.	PRODUKT / FOTO
S02 — ANTRIEBE					
11	CYL-31/32 + CYL-33/34	Doppelwirkender Magnet-Profilzylinder Ø80×100	2/2	2	 Luftec LDNT80X100-S Magnetic cushioned pneumatic cylinder Luftec
12	XV-20 / XV-21 / XV-24	Einfachwirkender Schwenkantrieb (Federrückzug)	2/2	3	 CONVALVE PAC Quarter-turn pneumatic actuator · spring return CONVALVE
S02 — SENSOREN / SCHALTER					
13	SWB20 / SWB21 / SWB24	Stellungsanzeige / Endschalterbox	2/2	3	 CONVALVE API200 ACTUATOR POSITION INDICATOR · 2×SPDT · IP67 CONVALVE
14	PPS03 / PPS04 / PPS07 / PPS08	Kolben-Magnetsensor (Reed)	2/2	4	 AIRTAC CMSE-07FY CMS Series · E groove · flying lead AIRTAC
UNBESCHRIFTET / ALLGEMEIN + ANGEBOTS-EXTRAS					
15	—	Schalldämpfer 3/8	1-2	1	 AN30-03 SILENCER 3/8 SMC
16	—	Schalldämpfer 1/2	1-2	1	 AN40-04 SILENCER 1/2 SMC
17	—	Modularer Zwischen-/Ersatzteil	—	2	 Y400T-D SPARE PART SMC
18	—	Modularer Zwischen-/Ersatzteil	—	1	 Y400-D SPARE PART SMC
ÜBERSICHT — KOLBENSSENSOREN					
19	PPS (alle)	Kolben-Magnetsensor — Einheitstyp	1-2	8	 AIRTAC CMSE-07FY All PPS items same product AIRTAC





Author : Fatih GÜRAL

Date : 01.10.2026

1326050-KSY-KBN 2B 1650

File : Pneumatic Diagram

Folio : 2/2

1 — Fonksiyon özeti

Endüstriyel parça yıkama makinesi KBN 2B 1650. Ana hatta hava hazırlanır; yön valfleri tahliye / dolum / kapak için silindir ve döner aktüatörleri sürer.

HAVA HAZIRLIK

HV-01 izolasyon → PR-01 filtre-regülatör (~6 bar) → V50 soft-start → PS06 basınç anahtarı.

5/2 VALFLER

Bir bobin bir yön, diğer bobin ters yön. V31/V32 ve V33/V34 her biri bir silindir.

3/2 + XV

V20 / V21 / V24, CONVALVE aktüatörleri (XV) sürer. SWB açık/kapalı teyidi verir.

GERİ BESLEME

PPS (CMSE-07FY) = silindir uçları. SWB/API200 = aktüatör kutu anahtarları.

2 — Tag ilişki haritası

PNÖMATİK TAG	SÜRER / OKUR	GÖREV
V31/V32	CYL-31/32 + PPS	Yıkama tahliye pistonu AÇ/KAPA (5/2)
V33/V34	CYL-33/34 + PPS	Durulama tahliye pistonu AÇ/KAPA (5/2)
V20 / V21 / V24	XV-20/21/24 + SWB	Dolum / kapak döner aktüatörler (3/2)
V50	Ana hat soft-start	Tüketicilerden önce yumuşak başlatma
V47	3/2 yardımcı	Tek bobin tüketici
V48/V49	5/3 tek gövde	Çift bobin orta egzoz
PPS...	CYL manyetik kanal	Piston reed sensörleri
SWB20/21/24	XV aktüatörler	OPEN/CLOSED mekanik switch

3 — Elektrik I/O (valf & aktüatörlü vanalar)

Omron CP2E · G9PS emniyet I/O · valf & aktüatör eşlemesi süreç fonksiyonuna göre

PLC / SAFETY I/O	ELEKTRİK FONKSİYONU	PNÖMATİK BAĞ (FONKSİYONEL)
Q1.1	YIKAMA TAHLİYE PİS. VALF AÇ	V31/V32 bobin A → CYL-31/32
Q1.2	YIKAMA TAHLİYE PİS. VALF KAPAT	V31/V32 bobin B → CYL-31/32
Q1.3	DURULAMA TAHLİYE PİS. VALF AÇ	V33/V34 bobin A → CYL-33/34
Q1.4	DURULAMA TAH. PİS. VALF KAPAT	V33/V34 bobin B → CYL-33/34
Q2.0	YIKAMA OTO. DOLUM VALFİ	3/2 dolum valfi (V20/V21/V24)
Q2.1	DURULAMA OTO. DOLUM VALFİ	3/2 dolum valfi (V20/V21/V24)
G9PS So0	COVER OPEN / Kapak aç	Kapak XV AÇ
G9PS So1	COVER CLOSE / Kapak kapat	Kapak XV KAPAT
G9PS So*	MAIN VALVE / Ana valf	Ana proses vanası
I1.3 / I1.4	YIKAMA TAHLİYE PİSTON AÇIK / KAPALI	CYL-31/32 PPS
I1.5 / I1.6	DURULAMA TAHLİYE PİSTON AÇIK / KAPALI	CYL-33/34 PPS
I2.3 / I2.4	YIKAMA OTO DOLUM SW BOX AÇIK / KAPALI	Dolum SWB
I2.5 / ...	DURULAMA OTO DOLUM SW BOX AÇIK / KAPALI	Dolum SWB
I2.6 / I2.7	COVER PISTON OPEN / CLOSED	Kapak konum geri bildirimi
I1.9 (via G9PS)	COVER CLOSED SIGNAL	Kapak kapalı emniyet kilidi

#	KOD	BİLEŞEN TIPI	FOLIO	ADET	ÜRÜN / FOTOĞRAF
S01 — ANA HAT / HAZIRLIK					
1	PR-01	Basınç regülatörü / Filtre-regülatör	1/2	1	 AW40-F04CE-D FILTER REGULATOR SMC
2	HV-01	3/2 yollu manuel valf (pressure relief)	1/2	1	 VHS40-F04-D PRESSURE RELIEF 3 PORT VALVE SMC
3	PS06	Modüler basınç anahtarı	1/2	1	 IS10E-40-F04-6L-D PRESSURE SWITCH MODULAR SMC
S01 — VALFLER					
4	V50	Yumuşak / yavaş başlatma valfi (soft-start)	1/2	1	 AV4000-F04-5DZB-A SOFT START VALVE 1/2-24VDC SMC
5	V47	3/2 yollu tek bobin solenoid valf	1/2	1	 AIRTAC 3V210-08 3/2 Single Solenoid · 1/4" AIRTAC
6	V48/V49	5/3 yollu çift bobin solenoid valf (tek gövde)	1/2	1	 AIRTAC 4V230E-08 5/3 Exhaust Center · 1/4" AIRTAC
S01 — AKTÜATÖRLER & SENSÖRLER					
7	CYL-48A / CYL-48B	Kilitli çift etkili silindir	1/2	2	 CP96NDB50-1200C CYLINDER WITH LOCK Ø50 STR=1200 SMC
8	PPS01A / PPS01B / PPS02A / PPS02B	Piston manyetik sensör (reed)	1/2	4	 AIRTAC CMSE-07FY CMS Series · E groove · flying lead AIRTAC
S02 — VALF MANIFOLD					
9	V20 / V21 / V24	3/2 yollu tek bobin solenoid valf	2/2	3	 AIRTAC 3V210-08 3/2 Single Solenoid · 1/4" AIRTAC
10	V31/V32 + V33/V34	5/2 yollu çift bobin solenoid valf	2/2	2	 AIRTAC 4V220-08 5/2 Double Solenoid · 1/4" AIRTAC

#	KOD	BİLEŞEN TIPI	FOLIO	ADET	ÜRÜN / FOTOĞRAF
S02 — AKTÜATÖRLER					
11	CYL-31/32 + CYL-33/34	Çift etkili manyetik profil silindir Ø80×100	2/2	2	 Luftec LDNT80X100-S Magnetic cushioned pneumatic cylinder Luftec
12	XV-20 / XV-21 / XV-24	Tek etkili döner aktüatör (yaylı)	2/2	3	 CONVALVE PAC Quarter-turn pneumatic actuator · spring return CONVALVE
S02 — SENSÖR / ANAHTAR					
13	SWB20 / SWB21 / SWB24	Aktüatör pozisyon göstergesi / limit switch box	2/2	3	 CONVALVE API200 ACTUATOR POSITION INDICATOR · 2×SPDT · IP67 CONVALVE
14	PPS03 / PPS04 / PPS07 / PPS08	Piston manyetik sensör (reed)	2/2	4	 AIRTAC CMSE-07FY CMS Series · E groove · flying lead AIRTAC
ETIKETSIZ / GENEL + TEKLİFTEKİ EK PARÇALAR					
15	—	Susturucu 3/8	1-2	1	 AN30-03 SILENCER 3/8 SMC
16	—	Susturucu 1/2	1-2	1	 AN40-04 SILENCER 1/2 SMC
17	—	Modüler ara / yedek parça	—	2	 Y400T-D SPARE PART SMC
18	—	Modüler ara / yedek parça	—	1	 Y400-D SPARE PART SMC
ÖZET — PISTON SENSÖRLERİ					
19	PPS (tümü)	Piston manyetik sensör — tek tip	1-2	8	 AIRTAC CMSE-07FY All PPS items same product AIRTAC